AI Equity Research Desk: four company calls, October 2026

Portfolio manager: Amanda. Six AI agents (Data, Risk, Fundamentals, Valuation, Bear case and Editor) each produced a buy/hold/sell memo on a listed company. The PM set every assumption, with a source for each one, and ruled on every challenge from the Bear case agent. All calculations were done by tested Python code, never by the AI itself. After the calls were made, the PM built her own DCF of Diageo in Excel (her formulas on the desk's sourced inputs); it reconciles to the desk's 1,780.2p.

Desk rule: BUY at +15% or more upside to the DCF fair value, SELL at -10% or worse, HOLD in between, with no overrides. Per-share values are in pence (GBp) for London listings and US dollars (USD) for Coca-Cola.

Company	Rating	Fair value	Share price	Upside	Bear case
Diageo plc (LSE: DGE)	**HOLD**	1,780.2p	1,673p (12 Oct 2026)	+6.4%	1,298.8p
The Coca-Cola Company (NYSE: KO)	**SELL**	USD 50.25	USD 90.12 (13 Oct 2026)	-44.2%	USD 37.86
Unilever PLC (LSE: ULVR)	**HOLD**	4,258.5p	4,710.5p (13 Oct 2026 close)	-9.6%	3,475.6p
RELX PLC (LSE: REL.L)	**SELL**	1,821p	2,771p (13 Oct 2026 close)	-34.3%	1,386p

The calls in one line (from each memo's thesis)

- **Diageo plc (LSE: DGE):** Fair value is a little above the price, and that rests on the discount rate.
- **The Coca-Cola Company (NYSE: KO):** The price needs a much lower cost of capital.
- **Unilever PLC (LSE: ULVR):** A HOLD by 19.1p that leans SELL.
- **RELX PLC (LSE: REL.L):** SELL on the desk's discount rate, not a firm call.

What is in this pack

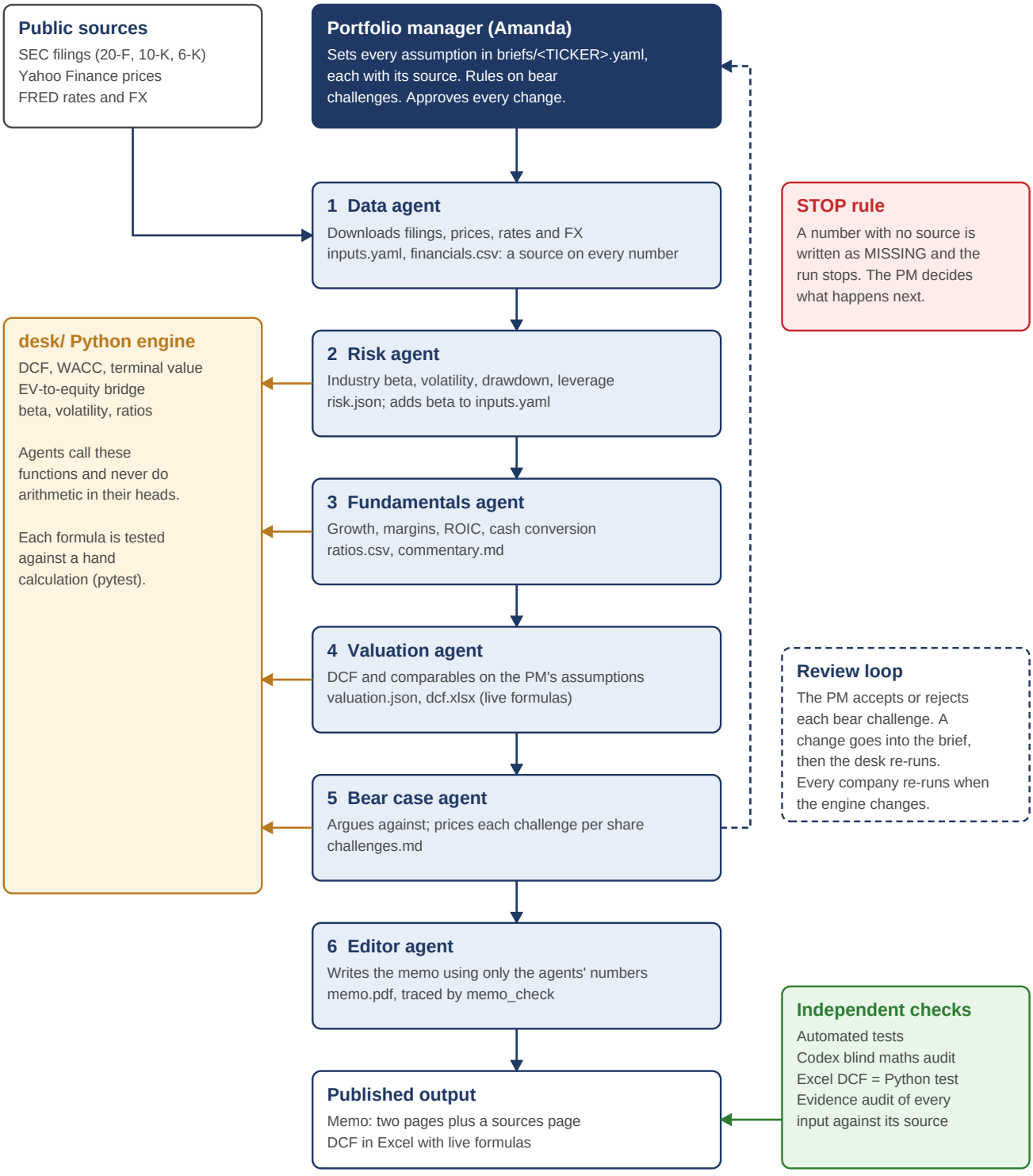
- **How the desk works:** a diagram of the six agents, the Python engine, the PM's review loop and the independent checks.
- **The four memos:** two pages each, plus a page of sources (SEC filings with page references, FRED series, data files).
- **What the agents got wrong:** the 34 mistakes logged during the project, how each was caught, and the rules that came out of them.

Read with care

- This is a student portfolio project, not investment advice. Prices are the closes of 12 and 13 October 2026.
- Three inputs could only be seen second-hand and are disclosed in the memos: the UK gilt and German Bund yields, and EABL's sales. The 4.2% equity risk premium (Damodaran implied) was confirmed on Damodaran's own site on 16 Oct 2026.
- Unilever is a borderline HOLD: it is 19.1p above the SELL line, and its memo shows the inputs that would flip it.

Code, agent outputs and evidence files: github.com/amandalombardo/ai-equity-research-desk

How the desk works

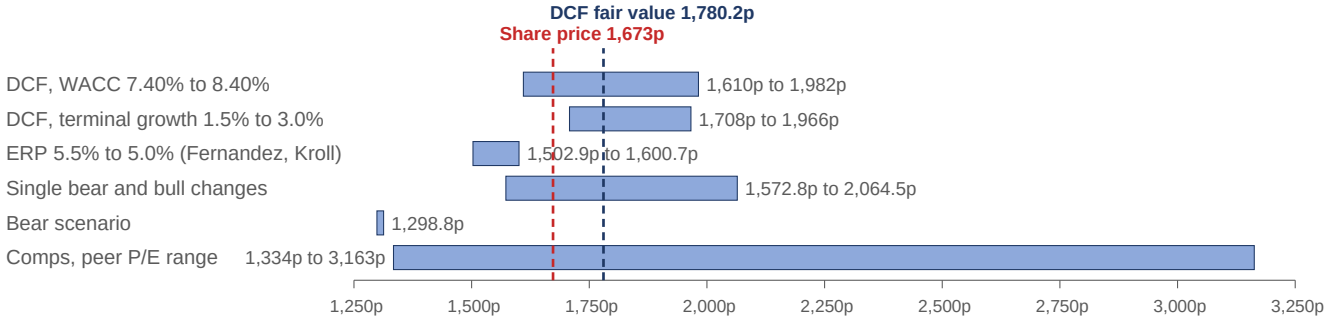


The agents run in order (/run-desk TICKER). Each writes only to its own folder and ends with notes.md: what it did, its flags, anything missing and its confidence. The Editor may use only numbers that trace back to an agent's output, which memo_check verifies.

Diageo plc (LSE: DGE)

HOLD	Fair value 1,780.2p	Share price 1,673p (12 Oct 2026)	Upside +6.4%	Bear case 1,298.8p

Desk rating: HOLD (PM confirmed; re-run 15 Oct 2026 after the evidence audit and source check). Rule: BUY at +15% or more, SELL at -10% or worse; no override. **SELL starts at 1,506p, 275p below fair value; BUY starts at 1,924p, 144p above it.** Per-share values are pence per ordinary share (GBp); the DCF is in USD m. Was 1,627.3p on 13 Oct: equity risk premium (ERP) 5.0% to 4.2% +183.5p; shares 2,224m -2.0p, then 2,231m diluted -6.8p; FY30-31 growth 2.0% (was 3.0%) -29.8p; beta 0.9526 from Damodaran's exact industry beta 0.7250 (was a rounded 0.73) +8.1p.



Thesis

- **Fair value is a little above the price, and that rests on the discount rate.** The DCF gives 1,780.2p (USD 23.56), 6.4% above 1,673p. The price implies a WACC of 8.20% (ours 7.90%), or an ERP of about 4.66% (ours 4.2%). At Kroll's 5.0% ERP the value is 1,600.7p, below the price; at the Fernandez survey's 5.5% it is 1,502.9p, a SELL by 3p.
- **Self-help is the whole profit story.** Organic net sales fell 2.0% in FY26 and North America fell 8.4%. The c.USD 1,000m savings plan drives profit growth; we assume 46% of it reaches profit, the low end of FY27-29 guidance.
- **The balance sheet limits the upside.** Leverage is 3.14x against a 2.5-3.0x target and the dividend has been cut. Disposal proceeds (USD 3,318m gross, tax MISSING) had not been received at 30 June 2026.

Valuation

DCF in USD m, valued at 30 June 2026 with mid-year discounting, rolled forward to 12 Oct 2026 at the cost of equity, converted at USD 1.3234 per GBP (FRED, 2 Oct 2026).

- **Growth path:** base net sales USD 18,505.5m (ex-EABL); organic growth 0.0%, 1.5%, 2.5%, 2.0%, 2.0% (FY27-31). FY30-31 was 3.0%, which had no source; the PM set 2.0%, the terminal rate, on 15 Oct 2026.
- **Savings flow-through:** 46% of cumulative savings (USD 377.5m, 962.5m, then 1,000m) reaches operating profit.
- **WACC 7.90%:** risk-free 5.22% (US 10-year, FRED, 8 Oct 2026), beta 0.95 (0.9526, industry bottom-up: Damodaran's global alcoholic beverages 0.7250 relevered at Diageo's D/E), **ERP 4.2%** (Damodaran implied ERP for the S&P 500, 1 Oct 2026; confirmed on Damodaran's site 16 Oct 2026; the old 5.0% had no source), cost of equity 9.22%; pre-tax cost of debt 6.22% (risk-free + 1%); debt weight 29.311%.
- **Terminal:** growth 2.0%; terminal capex 6.21% of sales (RONIC 13.4%, the FY26 ROIC). **Terminal value is 76.4% of EV** (above the desk's 75% flag); exit multiple 10.1x FY31 EBITDA vs c.9.78x at market today.
- **Minorities:** 2.566% x EV = USD 1,685m.

Bridge (USD m): EV 65,684 - net debt 20,482 - minorities 1,685 - restructuring cash 850 + associates 5,284 + disposal proceeds 3,318 (gross, tax MISSING) = equity 51,269 at 30 June; 52,561 at 12 Oct; 2,231m diluted shares = **1,780.2p**. No tax is deducted until Diageo discloses it (3.5p per USD 100m). The price is the 12 Oct close, **cum-dividend**: the 30c final dividend (about 22.6p) went ex on 15 Oct, so later prices are not like-for-like. ERP: SELL above 5.48%, BUY below 3.65%. Comps: median P/E 18.5x on four peers implies 2,311p (DCF 23.0% below); P/E only, two peers press-sourced, little weight.

Sensitivity of the DCF value (pence per share, GBp) to WACC and terminal growth (g); base in bold.

Value per share (p)	g 1.5%	g 2.0%	g 2.5%	g 3.0%
WACC 6.90%	2,104	2,226	2,375	2,562
WACC 7.40%	1,889	1,982	2,094	2,230
WACC 7.90%	1,708	**1,780**	1,865	1,966
WACC 8.40%	1,554	1,610	1,674	1,750

Value per share (p)	g 1.5%	g 2.0%	g 2.5%	g 3.0%
WACC 8.90%	1,420	1,464	1,514	1,572

Fundamentals

- **Growth has stalled:** organic growth was 6.5% (FY23), -0.6%, 1.7% and -2.0% (FY26). Africa (13.3%) and Latin America & Caribbean (7.7%) grew.
- **Margins compressed:** operating margin before exceptionals fell from 31.1% (FY22) to 28.9% (FY26).
- **Cash improved:** free cash flow rose from USD 2,235m (FY23) to USD 3,211m (FY26), helped by lower capex (6.1% of sales in FY26).
- **Returns fell:** ROIC went from 18.4% (FY23) to 13.4% (FY26); interest cover fell from 13.6x (FY22) to 7.0x (FY26).

Risks

- **Beta:** we use 0.95 (bottom-up), above all six regressions of Diageo's own shares (0.37 to 0.77). 0.72 gives 2,064.5p (BUY); BUY if beta is below 0.828, SELL above 1.244.
- **Volatility:** 31.6% (1 year) vs 12.5% for the FTSE 100.
- **Drawdown:** maximum -66.8%, from 4,098p (4 Jan 2022) to 1,362p (24 Mar 2026).
- **Leverage:** 3.14x adjusted net borrowings / adjusted EBITDA vs a 2.5-3.0x target. "About 2.6x after the disposals" is a floor: it is before tax on the proceeds and still counts EABL's EBITDA, which leaves with the sale, so true leverage is higher.

Bear case and our response

One single change now flips the rating to SELL: the Fernandez 5.5% ERP. No business challenge does on its own. All values below are pence (GBp). Responses are the PM's: her 13 Oct responses to challenges 1-6 were confirmed, and the open points decided, on 15 Oct 2026.

1. **Flow-through: most savings are reinvested.** 33% gives 1,733.7p, 25% gives 1,705.0p (HOLD). **Response:** 46% is the low end of FY27-29 profit guidance; we will check it against FY27 interim margins.

2. **Terminal returns: ROIC fell 5 points in three years.** RONIC 10.8% gives 1,708.5p; RONIC equal to WACC gives 1,572.8p (HOLD). **Response:** 13.4% is Diageo's lowest ROIC in four years, not a recovery, and FY26 capex (6.1% of sales) is close to the 6.21% the rule implies.

3. **No recovery: both 2023-26 incentive targets missed.** Zero organic growth gives 1,658.2p (HOLD). **Response:** the path is already the low end of guidance; five years of zero growth is our bear scenario, not the base. The unsourced 3.0% for FY30-31 is accepted as too high: it is now 2.0% (-29.8p).

4. **Terminal growth 2% vs falling spirits volumes.** 1.5% gives 1,708.2p (HOLD). **Response:** with 2.35% breakeven inflation, 2% nominal already means a small real decline; the IWSR figures are secondhand.

5. **Cost of capital.** Kroll's 5.0% ERP gives 1,600.7p (1,521.9p with Kroll's 20-year 5.60%, HOLD); the Fernandez 5.5% gives **1,502.9p (SELL)**; beta 1.00 gives 1,730.2p; a 2.0% credit spread 1,701.1p (HOLD). **Response:** our 1% spread is about the BBB average (1.02%, vs 0.69% for A-rated). ERP (PM, 15 Oct 2026): keep 4.2% (confirmed on Damodaran's site 16 Oct 2026), and show the Kroll value so readers can see that the upside depends on the ERP. The rating is HOLD at 4.2% and 5.0%, SELL at 5.5%.

6. **USL minority at market** (USD 5,778m with others at book): 1,638.1p (HOLD). **Response:** an upper bound: USL's minority at market but Diageo's 57.21% at the DCF multiple, and the USD 1,328m still includes EABL's 35%.

7. **Data inputs (audit).** Proceeds are gross (-3.5p per USD 100m of tax); the share count ignored dilution. **Response (PM, 15 Oct 2026):** diluted shares accepted and now in the base (2,231m, -6.8p). Proceeds are disclosed as gross, tax MISSING; nothing is deducted until Diageo discloses the tax.

What flips the rating. Alone, only the discount rate: SELL if ERP is above 5.48% or beta above 1.244; BUY if ERP is below 3.65% or beta below 0.828. Otherwise SELL needs a stack: the audit's bear lines with ERP 5.5% (1,445.8p) or with flow-through 33% and beta 1.00 (1,448.9p). A broad business bear at ERP 4.2% gives 1,608.6p, still HOLD but 3.8% below the price.

Open items (MISSING): peer EV/EBITDA and FCF yield, Constellation Brands EPS, calendarised peers, Moet Hennessy fair value, USL net cash, split of the USD 1,328m other minorities, tax on the disposals, UK or global implied ERP.

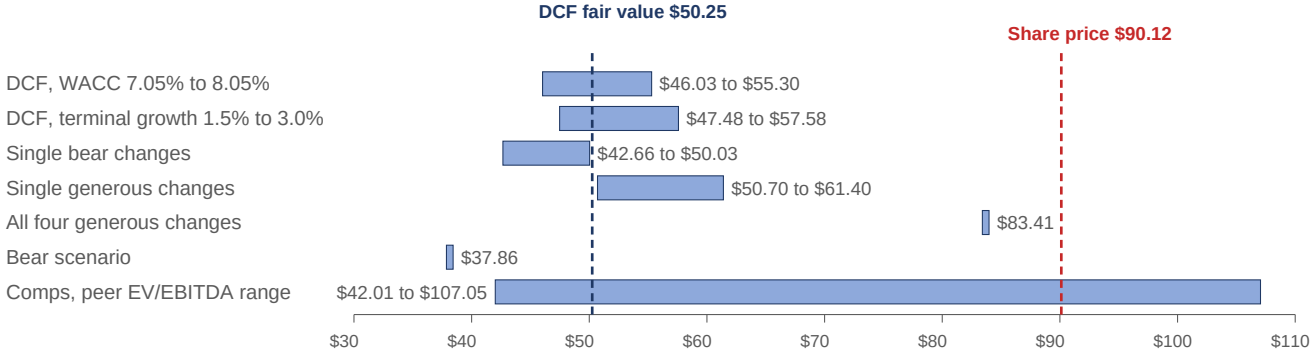
Sources

- Diageo Form 20-F FY26 (SEC accession 0001628280-26-057529): financial statements, p.18, p.22, p.85, p.110, p.164, p.186-189, p.210-211, regional reviews pp.27-31; 20-F FY22-FY25 (SEC EDGAR XBRL).
- Diageo interim results 6-Ks: FY26 (25 Feb 2026, accession 0001654954-26-001518) and FY25 (accession 0001654954-25-001149): first-half free cash flow.
- Diageo Capital Markets Day 6-K, 6 Aug 2026 (SEC accession 0001654954-26-007324): savings plan, guidance, restructuring cash.
- FRED: DGS10 (2026-10-08), DGS20 (2026-10-13), DEXUSUK (2026-10-02), T10YIE breakeven inflation and ICE BofA A, BBB and high-yield option-adjusted spreads (2026-10-08).
- Equity risk premium: Damodaran implied ERP for the S&P 500, 4.20% at 1 Oct 2026 (home page and ERPbymonth.xlsx, pages.stern.nyu.edu/~adamodar; read at source 16 Oct 2026); Kroll recommended US ERP 5.0%; Fernandez 2025 survey. Evidence pack docs/audit-2026-10-15/erp-evidence.md.
- Yahoo Finance (DGE.L in pence, FTSE 100, peer prices, USL price and shares, KES/USD, INR/USD); Damodaran, Betas by Sector (Global), file betaGlobal.xls dated 5 Jan 2026 (uploaded by the PM): Beverage (Alcoholic), 217 firms, unlevered beta corrected for cash 0.7250.
- Non-filing: EABL FY26 results, RCB sale, Pernod Ricard, Campari and Remy Cointreau results, IWSR (press/company). PM decisions: briefs/DGE.yaml; evidence audit docs/audit-2026-10-15/DGE.md and summary.md; evidence packs in docs/phase3/ and docs/phase4/. Agent outputs: runs/DGE/2026-10-12/.

The Coca-Cola Company (NYSE: KO)

SELL	Desk rating: SELL (PM confirmed; re-run 15 Oct 2026 after evidence audit)	Fair value USD 50.25	Share price USD 90.12 (13 Oct 2026)	Upside -44.2%	Bear case USD 37.86

Memo of 13 October 2026, re-run 15 October 2026 after the evidence audit (equity risk premium 4.2%, listed stakes no longer grown in the roll-forward, bear margin 31.9%) and the PM's bear responses (FY26 growth 5.0%, PM decision 15 Oct 2026). Desk rule: BUY at +15% or more, SELL at -10% or worse. **HOLD starts at USD 81.11, BUY at USD 103.64.** Per-share values in US dollars (USD); totals in USD m.



Thesis

- **The price needs a much lower cost of capital.** Our WACC (the yearly return lenders and shareholders require) is 7.55%; the USD 90.12 price implies 5.11%. HOLD would need 5.45%. No reasonable single change takes KO out of SELL.
- **The cash yield is below the risk-free rate.** FY26 free cash flow guidance of USD 12.4bn is a 3.20% yield on the market value, against 5.22% on the 10-year US Treasury, at about 30x earnings, before any IRS payment.
- **A high-quality business, fully priced.** ROIC is 27.0% and the margin ex-CCBA 33.2%. Only KO's own regression beta, an IRS win and 3% perpetual growth together reach HOLD (USD 83.41), still 7.4% below the price.

Valuation

DCF in USD m at the 31 Dec 2025 balance sheet, mid-year discounting (56% of FY26 guided cash, USD 6.9bn of USD 12.4bn, came in the first half).

- **Growth (USD, ex-CCBA):** base revenue USD 44,309m; 5.0% in FY26 (was 6.0%; PM decision 15 Oct 2026: part of the disposal headwind is not CCBA), then 3.0% a year (FY27-30).
- **Savings flow-through:** none (no announced programme); margin held flat at 33.2%. Tax 23.7%.
- **WACC 7.55%:** risk-free 5.22%, beta 0.6091, equity risk premium 4.2% (was 5.0%, unsourced), cost of equity 7.78%; cost of debt 5.91% (risk-free + 0.69% single-A spread); debt weight 7.1%. **ERP source:** Damodaran's implied ERP, 4.20% at 1 Oct 2026, confirmed at source (PM decision 15 Oct 2026).
- **Terminal:** growth 2.0%; capex 4.76% of sales, so new investment earns 27% (RONIC).

Bridge (USD m): EV 212,139 - net debt 29,686 - leases 1,722 - minorities 2,106 - IRS 14,000 + equity stakes 46,626 + CCBA proceeds 1,300 = 212,551; 216,169 at 13 Oct (listed stakes of USD 39,540m already at 13 Oct prices, so not grown; the rest grown at the cost of equity, less USD 6,840m of dividends); 4,302m shares = **USD 50.25**. **Terminal value is 76.7% of EV (FLAG, limit 75%).** Exit multiple 12.0x EBITDA, inside the peer range but at its bottom, against KO's own 26.1x. Comps: peer median EV/EBITDA 19.1x (PepsiCo 11.4x, Keurig Dr Pepper 19.1x, Monster 29.1x) gives USD 42.01 / 70.07 / 107.05; the DCF is 28.3% below the median. Little weight (three peers, forward P/E MISSING).

Value per share (USD)	g 1.5%	g 2.0%	g 2.5%	g 3.0%
WACC 6.55%	57.09	61.46	66.89	73.85
WACC 7.05%	51.85	55.30	59.49	64.71
WACC 7.55%	47.48	**50.25**	53.55	57.58
WACC 8.05%	43.78	46.03	48.69	51.86
WACC 8.55%	40.60	42.46	44.62	47.17

IRS case (PM decision: assume a loss). The bridge deducts USD 14,000m of tax and interest for 2010-25 (10-K), and tax is 19.9% + 3.8 points = 23.7%. **But the 8th Circuit reversed the 3M Tax Court ruling on the same rule (1 Oct 2025);** KO's 10-K calls it "highly supportive". KO's own appeal (11th Circuit, heard 25 Jun 2026) is pending. KO loses: **USD 50.25**; halfway: **USD 53.96**; KO wins: **USD 57.66** (USD 6.0bn deposit refunded, 19.9% tax). All SELL.

CCBA (African bottler being sold). The base removes CCBA (2024 revenue EUR 3,357m, EBIT EUR 246m) and adds the USD 1,300m sale price plus the retained 25% (USD 850m). The EUR figures are verified on slide 9 of Coca-Cola HBC's own presentation (FY2024 audited, excluding Ghana); slide 18 confirms USD 1.3bn for Coca-Cola's 41.52% and a USD 3.4bn equity value.

Fundamentals

- **Growth slowing:** organic 16% (FY21, FY22), 12% (FY23, FY24), 5% (FY25); reported 1.9% in FY25 after currency. Volume flat.
- **Margins up, partly mix:** comparable 28.7% (FY22) to 31.2% (FY25), helped by selling bottlers; impairments of USD 1,045m in FY25 are excluded.
- **Returns high:** ROIC 25.0%, 27.5%, 27.9%, 27.0% (FY22-25); 31.4% in FY25 without the IRS deposit.
- **Cash hit by one-offs, debt steady:** FY25 free cash flow USD 5,296m (USD 11,365m before the fairlife payment); net debt / comparable EBITDA fell from 2.40x (FY21) to 1.85x (FY25).

Risks

- **Beta:** 0.6091, the US soft-drinks industry beta (0.5755) relevered. KO's own regressions give 0.13 to 0.35, R-squared 1-12%. HOLD needs 0.059.
- **Volatility:** 18.9% (1 year) and 16.7% (5 years) vs 17.0% for the S&P 500 (5 years).
- **Drawdown:** maximum -17.3% over 5 years, USD 58.08 (22 Apr 2022) to USD 48.05 (5 Oct 2023), dividend-adjusted.
- **Leverage:** 1.85x net debt / comparable EBITDA (2.12x with leases); interest cover 8.3x; rated A+. Paying the USD 14,000m IRS liability would lift it to 2.72x.

Bear case and our response

No single bear change leaves SELL. Stacked bear (FY26 5.0%, 1% growth FY27-30, WACC +1 point, PM bear margin 31.9%): **USD 37.86**. Values in USD per share.

Challenge	Test: value (impact)	Response
1. Peak margin	30.0%: USD 45.15 (-5.09); 31.9%: USD 48.19 (-2.06)	30.0% includes CCBA. PM decision 15 Oct 2026: bear margin 31.9%, in the stacked bear
2. Currency drag	1% FY27-30: USD 46.70 (-3.54); FY26 5.0% instead of 6.0%: -0.51	PM decision 15 Oct 2026: FY26 5.0% accepted, in base and bear; 1% kept as a stress test
3. IRS and cash	add the USD 900m accrual: USD 50.03 (-0.22)	PM decision 15 Oct 2026: loss stays in base, no extra accrual; FY26 cash (USD 11,208m) reconciles
4. Stakes at market	31 Dec 2025 values: USD 49.03 (-1.22); 21% tax on gain: USD 49.00 (-1.25)	Small. PM decision 15 Oct 2026: keep no tax deducted
5. Beta at a low	0.868: USD 42.66 (-7.59), WACC 8.56%	PM decision 15 Oct 2026: keep 0.6091. The 8.56% is now inside the 6-9% range
6. ERP (audit)	5.0%: USD 46.54 (-3.71); 5.5%: USD 44.49 (-5.76)	PM decision 15 Oct 2026: keep 4.2% (confirmed); SELL at every ERP

Is the SELL just a harsh base case?

Desk assumption	Generous alternative	Value	Impact
Beta 0.61	0.35 (KO regression)	USD 61.40	+11.15
IRS case lost	Win, USD 6.0bn refunded	USD 57.66	+7.42
Terminal growth 2.0%	3.0%	USD 57.58	+7.33
RONIC 27.0%	31.4%	USD 50.70	+0.45
All four together		**USD 83.41**	**+33.16, HOLD**

PM view: SELL (PM decision 15 Oct 2026). Beta 0.35, an IRS win and 3% growth give USD 83.41 (USD 82.25 without RONIC), just above HOLD (USD 81.11) but 7.4% below the price, so the old line "every generous stack stays SELL" is withdrawn. Leave any one out and it is SELL (USD 66.17 to USD 75.01). None reaches BUY.

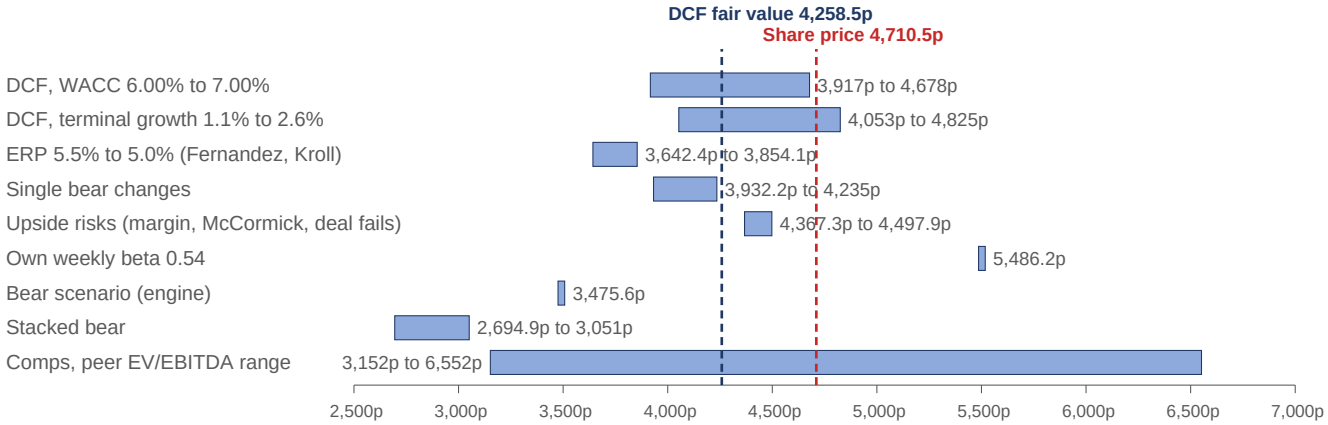
Sources

- Coca-Cola FY25 Form 10-K (SEC accession 0001628280-26-010047, filed 20 Feb 2026): balance sheet, Note 2 (held for sale), Note 10 (leases), Note 14 (pensions), Note 19, Note 20 (segments, pp.116-117), equity method investments note, IRS liability and the 3M 8th Circuit ruling (tax litigation note); 10-Ks FY22-FY24 (SEC EDGAR XBRL).
- Coca-Cola results 8-Ks (exhibit 99.1): Q4 2025 (0001628280-26-006642, pp.26, 28), Q2 2026 (0001628280-26-049922, FY26 outlook and first-half free cash flow), Q4 2019-Q4 2024 for the guidance track record.
- Coca-Cola 10-Qs: Q1 2026 (0001628280-26-028802) and Q2 2026 (0001628280-26-050503, Note 9: tax rate +3.8 points, USD 900m accrual).
- Coca-Cola HBC CCBA acquisition announcement, 21 Oct 2025 (presentation, slides 9, 13 and 18; PDF supplied by the PM and verified 15 Oct 2026. Slide 13's pro forma table shows EUR 3.43bn / EUR 0.253bn instead, about -USD 0.03 a share).
- Damodaran implied equity risk premium, 1 Oct 2026 (home page and ERPbymonth.xlsx, pages.stern.nyu.edu/~adamodar, read at source 16 Oct 2026; docs/audit-2026-10-15/erp-evidence.md). Kroll and Fernandez ERP figures from the same pack.
- FRED: DGS10 (2026-10-08), ICE BofA single-A spread BAMLC0A3CA (2026-10-08), DEXUSEU (2024 average).
- Yahoo Finance: KO, S&P 500, peer and listed-stake prices (13 Oct 2026), dividend dates; Damodaran, Betas by Sector (US), January 2026 (file supplied by the PM).
- Peers PepsiCo, Keurig Dr Pepper, Monster: SEC EDGAR XBRL. Evidence audit 15 Oct 2026: docs/audit-2026-10-15/KO.md and summary.md. PM decisions: briefs/KO.yaml. Agent outputs: runs/KO/2026-10-13/ (valuation and bear re-run 15 Oct 2026). PM bear responses: docs/audit-2026-10-15/bear-responses-2026-10-15.md (PM decision 15 Oct 2026, accept all).

Unilever PLC (LSE: ULVR)

HOLD	Fair value 4,258.5p	Share price 4,710.5p (13 Oct 2026 close)	Upside -9.6%	Bear case 3,475.6p

Desk rating: HOLD, borderline (PM confirmed 15 Oct 2026 after evidence audit; was SELL). Memo of 15 October 2026. Rule: BUY at +15% or more, SELL at -10% or worse. **SELL starts below 4,239.45p, only 19.1p (0.4% of the price, about EUR 486m of value) under fair value; BUY starts at 5,417.1p.** The PM kept the desk rule rather than override to SELL, so all four companies are rated by the same rule. **The call leans SELL:** each of the five bear challenges alone, Kroll's 5.0% ERP (3,854.1p), the 9 Oct Bund of 3.50% (4,222.4p) and a 1.0% credit spread (4,231.8p) each make it SELL. Only Damodaran's January ERP of 4.23% keeps HOLD (4,241.6p). Per-share values in pence (GBp); totals in EUR m (Unilever reports in euros), at EUR 1.18190 per GBP; 2,154.3m shares.



Thesis

- **A HOLD by 19.1p that leans SELL.** The move from 3,855p (SELL) to 4,258.5p comes from five PM decisions of 15 Oct 2026; the ERP cut from 5.0% to 4.2% gives +379.1p of the +403.3p. The ERP, the Bund and the McCormick price each decide the rating within a few basis points or 2.4%.
- **The price already assumes a best case for the business that remains.** At 4,710.5p the market pays 14.7x EBITDA for Unilever without Foods, against a peer median of 11.9x and a DCF exit of 12.2x. The comps median (3,902p, -17.2%) says SELL; the DCF says HOLD.
- **About a third of the value is the McCormick deal at today's price, untaxed.** Foods consideration 1,308p (31%). McCormick is USD 44.64, 23% below the USD 57.84 deal reference; USD 43.55 makes it SELL. Upside risks: a failed deal (+115.0p to +150.5p), margin 18.8% (4,497.9p), own beta 0.54 (5,486.2p, BUY).

Valuation: a sum of the parts

As if Foods were already sold: a DCF of Home and Personal Care less outside shareholders, plus McCormick's payment for Foods at today's price, plus stakes and JVs, less net debt, pensions and other claims.

- **Growth (EUR, ex-Foods):** base sales EUR 39,803m; **+2.0% in FY26** (H1 2026 change of the three businesses valued), then 1.6% a year FY27-30.
- **Margin:** 17.7% flat (ex-Foods 18.8% less 1.1% recurring restructuring). Savings flow-through: none modelled. Tax 26.0%; capex 3.15%, D&A 2.6% of sales.
- **WACC 6.50%:** Bund 3.45%, beta 0.8844, **ERP 4.2%** (Damodaran implied, 1 Oct 2026), cost of equity 7.164%; cost of debt 4.15% (Bund + 0.70%), 3.071% after tax; debt weight 16.13%. Market-implied WACC 5.97%.
- **Terminal:** growth 1.6%, RONIC 17.8%. Terminal value 77.9% of EV (FLAG, limit 75%); exit 12.2x EBITDA.

Bridge (EUR m, 31 Dec 2025): EV 103,329 - minorities 11,040 (10.684% of EV) - net debt 23,076 - restructuring 500 - pension deficit after tax 698.6 + stakes and JVs 7,533.5 + Foods consideration 33,312.4 = 108,861; rolled to 108,431 at 14 Oct 2026 = **EUR 50.33 = 4,258.5p**. Parts (pence): DCF after minorities 3,827; Foods 1,308; other stakes 308; net debt, restructuring, pension, dividends and buyback -1,185.

Step (pence)	Change	Value	Upside	Rating
14 Oct run		3,855.3	-18.2%	SELL
1. ERP 5.0% to 4.2%	+379.1	4,234.4	-10.1%	SELL
2. Pension deficit	-29.0	4,205.4	-10.7%	SELL

Step (pence)	Change	Value	Upside	Rating
3. MKC dividends	-25.2	4,180.2	-11.3%	SELL
4. JV at new Ke	+19.6	4,199.7	-10.8%	SELL
5. FY26 growth 2.0%	+58.8	**4,258.5**	**-9.6%**	**HOLD**

HOLD turns SELL at	Used	Breakeven
ERP / Bund (risk-free)	4.20% / 3.45%	4.234% / 3.476%
FY26 growth / terminal growth	+2.0% / 1.6%	+1.51% / 1.560%
McCormick price / margin / beta	USD 44.64 / 17.7% / 0.8844	USD 43.55 / 17.61% / 0.892
Foods tax (39.3p per EUR 1bn)	not deducted	above about EUR 0.49bn

Pence	g 1.1%	g 1.6%	g 2.1%	g 2.6%
WACC 6.00%	4,410	4,678	5,012	5,443
WACC 6.50%	4,053	**4,258.5**	4,510	4,825
WACC 7.00%	3,755	3,917	4,111	4,348

Comps: peer median EV/EBITDA 11.9x (7 HPC peers; four on Yahoo data) gives 3,152p / 3,902p / 6,552p; without KMB 4,575p (HOLD).

Disclosures. (1) ERP 4.2% confirmed on Damodaran's site 16 Oct 2026 (Damodaran's 1 Jan figure, 4.23%, also confirmed). (2) Bund 3.45% from aggregators only. (3) The 0.70% spread is a US-dollar index added to a euro yield. (4) Terminal value 77.9% of EV. (5) Comps say SELL, the DCF says HOLD. (6) **MISSING:** Foods tax; Foods interim cash flows to closing (indicator about 113p, not used). (7) McCormick dividends for 3 quarters; 2 quarters gives 4,267.0p. (8) FY26 +2.0% is an H1 figure. Foods cash received mid-2027 costs 15.7p (HOLD by 3.5p); the whole consideration at WACC 57.2p (SELL); neither is in the base.

Fundamentals

- **Weak euro growth:** USG 4.5%, 9.0%, 7.0%, 4.2%, 3.5% (FY21-25), but turnover fell 3.8% in euros in FY25; 62% of the remaining company is in emerging markets.
- **Margins at a peak:** underlying 18.4%, 16.1%, 17.6%, 19.4%, 20.0% (FY21-22 include Ice Cream, FY23-25 do not); restructuring recurs at 0.8-1.4% of turnover.
- **Returns and mix:** ROIC 19.0% (FY25), 17.8% after restructuring (ex-Foods not disclosed). Margins: Personal Care 22.6%, Beauty & Wellbeing 19.2%, Home Care 14.9%; Foods perimeter 24.3%.

Risks

- **Beta:** 0.8844 bottom-up (Damodaran global Household Products 0.7741, relevered at D/E 0.1924). Own regressions vs the FTSE 100: 0.43, 0.54, 0.91 (R-squared 0.08-0.30). SELL at 0.892; 0.91 gives 4,191.0p; June 2026 net debt gives 0.8981.
- **Volatility and drawdown:** 22.4% (1 year) and 19.2% (5 years) vs 11.6% and 12.5% for the FTSE 100; McCormick 30.5%. Drawdown -25.5% over 5 years, 5,470.6p (24 Feb 2026) to 4,077.7p (4 Jun 2026).
- **Leverage:** net debt / underlying EBITDA 2.03x (Dec 2025) and 2.27x (Jun 2026) vs a c.2x target; rated A+/A1.

Bear case and our response

Pence per share; impact vs 4,258.5p. Each challenge alone turns HOLD into SELL. Stacked bear (engine bear + JV fix + 16.2% margin + 25% tax): 3,051.0p; with HUL at market 2,694.9p.

Challenge	Test: value (impact)	PM decision 15 Oct 2026
U1. Foods consideration worth less	Tax 25% of cash: 4,120.8 (-137.7); McCormick USD 33.15: 4,057.2 (-201.4); deal fails +115.0 to +150.5	Keep method (no tax until sourced); "stays SELL" withdrawn; tax above EUR 0.49bn or MKC -2.4% gives SELL
U2. 17.7% margin is a peak	16.2%: 3,932.2 (-326.3)	Keep 17.7%; "stays SELL" withdrawn
U3. HUL minorities at market	4,087.1 (-171.4) to 3,997.4 (-261.2)	Keep proportionate (Diageo rule); market value is a rating risk
U4. Growth and terminal value	g 1.1%: 4,037.5 (-221.1); RONIC 16.5%: 4,235.0 (-23.6); RONIC = WACC: 3,747.3	Keep 1.6%, partly accepted; "stays SELL" withdrawn
U5. Discount rate	Beta 0.9998: 3,971.9 (-286.7); ERP 5.0%: 3,854.1 (-404.4)	Keep beta 0.8844, ERP 4.2% (desk-wide); "SELL either way" withdrawn
U6. Overall view	Engine bear 3,475.6; JVs at Ke + 1 pt 3,449.2	Withdrawn: HOLD by 19.1p; each challenge alone gives SELL; upside: failed deal, higher margin, lower beta

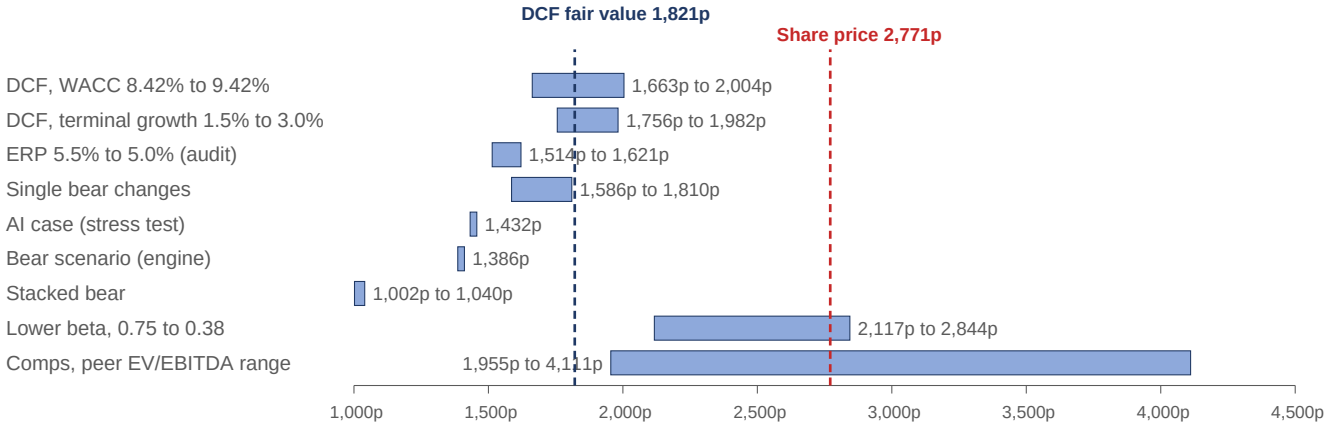
Sources

- Unilever FY25 Form 20-F (SEC accession 0000217410-26-000007, filed 12 Mar 2026): balance sheet p.131 (pension plans in deficit), cash flow p.132, notes 2 (pp.137-138), 4B, 11 (p.158), 13-14 (pp.159-160), 15B-15C (pp.162-164), 17A (p.174), related undertakings p.194; XBRL.
- Unilever FY25 results 6-K (0001654954-26-001155, 12 Feb 2026); H1 2026 results 6-K (0000217410-26-000057, 28 Jul 2026; HPC business-group turnover); FY17-FY24 results 6-Ks (guidance record); Total Voting Rights 6-Ks (0001654954-26-008017, 0001654954-26-007194).
- Foods/McCormick Form 425 investor presentation (0000217410-26-000027, 31 Mar 2026) and call transcript (0000217410-26-000032); McCormick 10-Q to 31 Aug 2026 (0000063754-26-000329: note 2, quarterly dividend); TMICC Form 424(b)(4) (0001104659-26-042683, 13 Apr 2026).
- ERP: Damodaran implied ERP (pages.stern.nyu.edu/~adamodar, read at source 16 Oct 2026); Kroll and Fernandez et al. 2025 seen in search summaries only (docs/audit-2026-10-15/erp-evidence.md). Bund: Trading Economics and Bloomberg rates page (aggregators); 9 Oct cross-check thetrading.tools. FRED: BAMLC0A3CA, DEXUSEU, DEXUSUK, DEXINUS (9 Oct 2026). Damodaran betaGlobal.xls (5 Jan 2026, PM upload).
- Yahoo Finance (non-filing): ULVR.L, ^FTSE, MKC, MICC.AS, HINDUNILVR.NS and peer prices (13 Oct 2026). Peers: SEC EDGAR XBRL (PG, CL, KMB); Yahoo (Reckitt, Beiersdorf, L'Oreal, Henkel).
- PM decisions: briefs/ULVR.yaml; docs/phase5/brief-unilever.md; evidence audit docs/audit-2026-10-15/ULVR.md and summary.md; bear responses docs/audit-2026-10-15/bear-responses-2026-10-15.md (U1-U6). Agent outputs: runs/ULVR/2026-10-14/ (data, risk, fundamentals, valuation, bear).

RELX PLC (LSE: REL.L)

SELL	Desk rating: SELL (PM confirmed 15 Oct 2026; re-run 15 Oct 2026 after evidence audit)	Fair value 1,821p	Share price 2,771p (13 Oct 2026 close)	Upside -34.3%	Bear case 1,386p

Memo of 14 Oct 2026, updated 15 Oct 2026 after the evidence audit: ERP now 4.2% (Damodaran implied, 1 Oct 2026; was 5.0%, no source); nothing else in the base changed. **HOLD starts at 2,494p (-10%), BUY at 3,187p (+15%).** Per share in pence (GBp); totals in GBP m (no exchange rate); 1,743.75m shares (25 Sep 2026).



Thesis

- **SELL on the desk's discount rate, not a firm call.** The DCF gives 1,821p, 34.3% below the price, at a WACC of 8.92%, now inside the desk's 6-9% range but only 0.08 pt below the top. Cost of equity is 9.54% (gilt 5.45%, beta 0.9729, ERP 4.2%). On these cash flows the price implies a WACC of 6.93%.
- **The discount rate alone can flip it.** HOLD needs a WACC of 7.38%, a beta of 0.535 or a gilt of 3.80%. The DCF is 39.1% below the comps median (2,990p), and its exit multiple of 8.6x EBITDA is below every peer (11.1x to 20.7x; RELX 14.6x) (FLAG). RELX's own 5-year monthly beta (0.3783) gives 2,844p, HOLD (+2.6%), but its R-squared is 0.04.
- **The cash flows cannot flip it.** HOLD would need a 44.5% margin, 12.4% a year growth in FY27-30 or 4.89% terminal growth. Every sourced bear and audit line stays SELL.

Valuation

- **Growth (GBP):** base sales GBP 9,590m (FY25, incl. print 399); +3.0% in FY26, then 5.0% a year FY27-30 (constant-currency, already net of print run-off and disposals). Independent industry growth forecast: MISSING (sources blocked).
- **Margin:** 35.0% flat (H1 2026, without print); savings flow-through none. Tax 22.8%; capex 7.10%, D&A 5.07%, working capital -15.5% of sales.
- **WACC 8.92%:** UK gilt 5.45% (13 Oct 2026, aggregator sites only; Bank of England blocked here, FLAG), beta 0.9729 (Damodaran global Information Services 0.8737, relevered), ERP 4.2% (Damodaran implied, 1 Oct 2026, confirmed at source 16 Oct 2026; PM decision 15 Oct 2026: keep), cost of equity 9.54%; cost of debt 6.15% (gilt + 0.70% single-A spread, a US-dollar index added to a sterling gilt, FLAG), 4.75% after tax; debt weight 12.82%.
- **Terminal:** growth 2.0%, RONIC 15.4%. Terminal value is 71.5% of EV; implied exit 8.6x EBITDA.

Bridge (GBP m, 31 Dec 2025): EV 39,616.5 - net debt 7,104 (ex leases) - minorities 24 + investments 131 = 32,619.5; rolled forward at 9.54% to 13 Oct 2026, less dividends 1,217 and buybacks 2,056.2 = 31,759.4 = **GBP 18.21 = 1,821p**. Audit disclosures: Employee Benefit Trust shares are in the count (+5.6p if excluded, per the audit); the GBP 366m interim dividend is RELX's expected cost, not yet published cash; buybacks after 25 Sep 2026 are not yet filed (cash and shares both left out).

Value per share (pence)	g 1.5%	g 2.0%	g 2.5%	g 3.0%
WACC 7.92%	2,116	2,218	2,338	2,482
WACC 8.42%	1,923	2,004	2,098	2,209
WACC 8.92%	1,756	**1,821**	1,896	1,982
WACC 9.42%	1,611	1,663	1,722	1,790

Value per share (pence)	g 1.5%	g 2.0%	g 2.5%	g 3.0%
WACC 9.92%	1,482	1,525	1,572	1,626

All SELL (top corner 12p short of HOLD). Comps at 15.7x EV/EBITDA on GBP 3,902m: 1,955p / 2,990p / 4,111p; four peers Yahoo-only; NTM EBITDA MISSING.

Fundamentals

- **Steady underlying growth, weak in pounds:** underlying +7% in FY24, FY25 and H1 2026; reported in GBP +3.0% and +1.7% (FY24-25), held back by currency, print run-off (-22.8% in FY25) and disposals.
- **Margins rising every year:** adjusted 30.5% (FY21) to 34.8% (FY25) and 35.5% in H1 2026; without print 32.4% (FY23) to 35.0% (H1 2026). Flat 35.0% is cautious (71p a point). ROIC 11.9% rising to 15.4%; cash conversion 97-101%.
- **Mix (FY25 underlying growth):** Risk +8%, STM +5%, Legal +9% (+10% in H1 2026), Exhibitions +8%.

Risks

- **Beta:** 0.9729 bottom-up. Own regressions vs the FTSE 100: 0.67 (2y weekly), 0.66 (5y weekly), 0.38 (5y monthly), R-squared 0.04-0.12; peer median 0.75. Main risk: HOLD needs 0.535, near the Blume 0.58 (2,400p).
- **Volatility and drawdown:** 36.0% (1 year) vs 11.6% for the FTSE 100; largest daily fall -14.4% (3 Feb 2026). Drawdown -51.3% over 5 years, 4,133p (13 Feb 2025) to 2,012p (11 Feb 2026), dividend-adjusted (FTSE 100: -13.4%).
- **Leverage:** net debt / adjusted EBITDA 1.87x (Dec 2025), 2.22x (Jun 2026, after GBP 1,750m of buybacks); no target; A3 / A- / BBB+. More debt slightly raises the DCF (+13p): a risk, not a value line.

Bear case and our response

Pence, impact vs 1,821p (ERP 4.2%). A: against buying; B: against the SELL. **PM decision 15 Oct 2026:** all responses kept (bear-responses-2026-10-15.md, R1-R4). **PM view:** SELL on the desk's discount rate, moderate conviction; the call rests on today's gilt and a market-average beta; gilt trigger for HOLD 3.80%.

Challenge	Test at ERP 4.2%: value (impact)	PM response (PM decision 15 Oct 2026)	Still holds?
A1. Bear margin 33.1% incl. print	32.4%: 1,636p (-185p)	Accepted: bear margin is now 32.4% (like-for-like)	Yes; SELL
A2. AI caps Legal and STM pricing	Growth 3%: -160p; AI case: 1,432p (-389p)	Main risk, not a base change: Legal growth is speeding up and base growth is already below organic	Yes; AI case SELL
A3. Reinvestment under-charged	RONIC = WACC -189p; acquisitions -73p; tax 25.0% -65p	Keep: RELX's own ROIC and tax estimate; growth is organic, so acquisitions stay out	Yes; all SELL
A4. Discount rate, leverage, price path	Engine bear 1,386p; beta 1.195 -235p; ERP 5.0% / 5.5%: 1,621p / 1,514p	Keep latest Damodaran beta; keep ERP 4.2% (confirmed at source 16 Oct 2026)	Yes; all SELL
A5. Stacked bear	1,002p (-819p)	Noted; SELL	Yes; SELL
B1. Market-implied WACC 6.93%	Needs beta 0.429	Rejected as a basis: needs a beta below every peer	Yes; below every peer (0.58)
B2/B8. 2.0% terminal growth vs 5.45% gilt; other gilts	+48p to +174p; gilt 5.30% / 5.21%: +48p / +78p	Partly accepted: the mismatch is real but small; gilt on the run date per the desk rule	Yes; all SELL
B3. Beta 0.973 vs regressions and peers	0.75: 2,117p; 0.58: 2,400p; 0.38: 2,844p (HOLD)	Keep 0.973: HOLD would need 0.535; even 0.75 gives 2,117p SELL; lower-beta generous stacks reach HOLD (2,546p / 2,586p)	Yes; at 0.973 the best generous stack is 2,409p
B4. Implied exit 8.6x below every peer	g 4.38% for 11.1x: 2,314p	Noted: a symptom of the discount rate	Yes; same reading
B5. Flat margin vs rising record	35.5%: +35p	Keep flat: no number from RELX; cautious and stated	Yes; SELL
B6. 5% growth vs 7% underlying	7% FY27-30: +169p	Keep: 7% leaves out print run-off and disposals	Yes; SELL
B7. Comps 2,990p	+7.9% vs price, HOLD territory	Cross-check only: peers fell on the same AI fears	Yes; DCF now 39.1% below

Sources

- RELX Annual Report 2025 (20-F ex 15.2, SEC accession 0001104659-26-017277, filed 19 Feb 2026): balance sheet p.141, notes 1-2 (pp.145-147), note 15 (p.167), note 20 (p.176), five-year summary p.186, p.199, pp.204-206; XBRL.
- RELX FY2025 results 6-K ex 99.1 (0000929869-26-000008, 12 Feb 2026) pp.2-29; H1 2026 interim results 6-K ex 99.1 (0000929869-26-000054, 23 Jul 2026) pp.1-28; buyback, dividend and voting-rights 6-Ks (0000929869-26-000041, -000056, -000064, -000066, -000068, -000071); FY21-FY24 results 6-Ks (guidance record); RELX Capital term sheet (0001193125-25-061449, 24 Mar 2025).
- UK 10-year gilt on 13 Oct 2026: Investing.com and Trading Economics; aggregators, not a primary source (Bank of England site blocked). FRED: BAMLC0A3CA, BAMLC0A4CBBB (US-dollar corporate indices), DGS10, DGS20, DFII10, T10YIE, IRLTLT01GBM156N. Damodaran betaGlobal.xls (5 Jan 2026, PM upload).
- Equity risk premium 4.2%: Damodaran implied ERP, 1 Oct 2026 (PM decision 15 Oct 2026; docs/audit-2026-10-15/erp-evidence.md; read at source 16 Oct 2026, pages.stern.nyu.edu/~adamodar). Alternatives: Kroll 5.0%, Fernandez et al. 2025 survey 5.5% (SSRN 5260463).
- Evidence audit 15 Oct 2026: docs/audit-2026-10-15/RELX.md and summary.md (findings V1, V2, M2, M3, C2, C5; EBT impact +5.6p).
- Yahoo Finance (non-filing): REL.L, ^FTSE and peer prices (to 13 Oct 2026). Peers: SEC EDGAR XBRL (TRI, VRSK, SPGI, MCO); Yahoo (Wolters Kluwer, Experian, LSEG, Informa). Thomson Reuters Q2 2026 6-K (0001193125-26-333835).
- PM decisions: briefs/RELX.yaml, docs/phase5/brief-relx.md and docs/phase5/relx-*.md (bear responses: relx-bear-responses.md); PM decision 15 Oct 2026 on bear responses: docs/audit-2026-10-15/bear-responses-2026-10-15.md (items R1-R4, "Accept all"). Agent outputs: runs/RELX/2026-10-14/ (data, risk, fundamentals, valuation, bear; re-run sections of 15 Oct 2026).

What the AI agents got wrong, and how we caught it

AI Equity Research Desk · write-up of the mistakes log, 12 to 16 October 2026 · Amanda (portfolio manager)

Six AI agents valued four companies (Diageo, Coca-Cola, Unilever and RELX) in four days. Every error we found was written down as it happened, in a mistakes log of 34 rows (docs/mistakes-log.md, with the row numbers used below). This note is what that log teaches. Per-share values are in pence (GBp) for the three London-listed companies and in US dollars (USD) for Coca-Cola.

The short version

- **The AI never got a sum wrong in its head, because it was never allowed to do one.** All arithmetic lives in tested Python code (the desk/ folder). The errors were all of a different kind: an assumption with no source, a number that meant something else, two items measured on different dates or over different businesses, or a rule that was changed in one place but not another.
- **The dangerous errors looked reasonable.** A sales recovery to 3 to 4% (row 1), a 5.0% equity risk premium carried from company to company (row 27) and a beta copied from a rounded search snippet (row 33) were all plausible. None had a source. Plausibility is not evidence.
- **The checks that worked were the ones built into the process.** The rule "every number needs a source, or the agent stops" and a full evidence audit caught errors that an independent maths audit had passed.
- **One error changed a rating and another nearly did.** Diageo's first run said BUY at 2,475p on a beta and cost of debt that were mechanically correct but economically meaningless (row 7). Unilever crossed from SELL to HOLD only when, on top of the lower risk premium, its growth rate was measured on the same businesses as the forecast (row 32).

Kinds of mistake

Kind of mistake	Rows	Example
A plausible number with no source	1, 27, 33	The 5.0% equity risk premium was copied into all four companies as a "portfolio rule"; each 0.5 point moved Unilever by about 220p
Like-for-like: comparing different sets of businesses or assets	14, 16, 23, 26, 32, 34	Unilever's FY26 growth (+0.5%) included Foods, which the model had already removed; the businesses actually valued grew +2.0%
Dates: items measured at different points in time	5, 12, 15, 17, 22, 24	Unilever's share count was after a EUR 1,507m buyback but the cash was never deducted: about 59p too high
Labels and units: a number that meant something else	3, 4, 6, 18, 25, 29	Diageo's "Revenue" tag includes excise duty (about USD 28bn) rather than net sales (about USD 20bn)
A correct formula giving a meaningless answer	7, 9, 20	Unilever's euro cash flows were first discounted with a UK gilt yield (4.99%) rather than the German Bund (3.18%): about 1.8 points of WACC
A rule changed in one place but not everywhere	2, 8, 10, 13, 28	A retired script could have overwritten Diageo's beta and turned SELL into BUY (+694p) if anyone had re-run it
A disclosure that was missed	11, 19, 21, 30	Diageo's c.USD 850m of restructuring cash for FY27-29 appeared nowhere in the model (about -30p)
Process	31	An audit sub-agent sent the PM's email address to the SEC as its contact header

Some rows fit two kinds; each is counted once, under its main cause.

Who caught them

Caught by	Rows	Count
The builder's own reviews and tie-outs to filings (Claude Code)	3, 4, 5, 6, 8, 9, 11, 12, 13, 14, 15, 16, 17, 19, 20, 21	16
Another agent on the desk, or the agent itself	7, 18, 22, 23, 24, 25, 26, 31	8

Caught by	Rows	Count
The evidence audit and source check the PM ordered on 15 Oct, and the check at source on 16 Oct	27, 28, 29, 30, 32, 33, 34	7
The PM refusing an unsupported assumption	1	1
An automated test	2	1
Codex, the independent maths auditor	10	1
Total		**34**

Two things stand out. First, the agents checked each other: the Bear case agent rejected a beta of 0.37 with an R-squared of 0.04 (row 7), the Data agent noticed that net debt and the share count were at different dates (row 22), and the Data agent stopped the RELX run rather than guess when its download failed (row 25). Second, the independent maths audit was useful but not enough. Codex found the retired beta script (row 10) and independently confirmed the terminal-value error (row 9), but it missed the minority-interest double count (row 14) and the missing restructuring cash (row 11). Code that computes the right formula can still be fed the wrong number. Only checking each input against its source finds that.

The five mistakes that mattered most

1. Correct maths, wrong answer (row 7, Diageo). The first run calculated a cost of debt of 3.81%, below the 5.22% risk-free rate, and a five-year beta of 0.37 with an R-squared of 0.04. Both were arithmetically right and economically meaningless: no lender charges Diageo less than the US government, and a beta that explains 4% of the share's moves is noise. The result was BUY at 2,475p. The PM set the cost of debt to the risk-free rate plus 1% and asked for an evidence-based beta. Of seven estimates tested, she chose the industry bottom-up beta. The rating became SELL at 1,448p.

2. A default is still an assumption (row 27, all four companies). The 5.0% equity risk premium (the extra return investors demand for owning shares rather than government bonds) was set for Diageo and copied into every later brief. It had no source. The PM replaced it with Damodaran's implied premium of 4.2% (1 October 2026), because that measure is built from today's market prices, like the bond yields the desk uses. All four companies were re-run. This premium still decides close calls: at 5.5% Diageo becomes a SELL by 3p.

3. Same businesses on both sides (rows 16 and 32). For Coca-Cola, the proposed growth guidance already included the sale of the African bottler, which the model had also removed from the base year, so the disposal would have been counted twice. The same mistake then happened again for Unilever: growth included the Foods business being sold, while the model valued only the remaining businesses. Together with the lower risk premium, fixing it moved Unilever from SELL to HOLD (4,258.5p against 4,710.5p). The lesson was written down after the first time and still repeated, which is why it is now a check on every company.

4. Every item in the bridge needs a date (rows 12, 17, 22 and 24). Going from the value of the business to the value per share means adding and subtracting many items: debt, dividends, buybacks, stakes, legal liabilities. Each one is measured on some date. Twice an item was counted twice because of dates: Coca-Cola's 2026 tax charge (row 17), and listed stakes already priced on the run date but grown again (row 24). Once a cash outflow was missed: Unilever's buybacks (row 22). And once a dividend was nearly deducted although the share price still included it (row 12).

5. Copy from the source, not from a summary (row 33). Diageo's industry beta was typed as 0.73 from a rounded search result instead of 0.7250 from the Damodaran file itself. The value moved from 1,772.1p to 1,780.2p, with no change of rating. Small, but it is the clearest example of why the desk now treats a search result only as a pointer to the real source.

How the rules changed

Each mistake became a rule in the desk's instructions, the code or the briefs:

- Every assumption has a source in the brief, or it is written as MISSING and the run stops.
- A decision by the PM lives in a field the agents read, never only in a comment (row 28).
- When the engine changes, every company it affects is re-run (row 28).
- Growth, margins, capex and depreciation must cover the same businesses and assets as the base year (rows 16, 23, 32).
- Every item in the equity bridge carries a date, and only items at the balance-sheet date are rolled forward (rows 22, 24).
- The discount rate follows the currency of the cash flows, not of the share price (row 20).
- Inputs are copied at full precision from the source file (row 33).

- Each change of method gets an automated test, including values exactly on a rating threshold (row 2).

How the ratings moved

Diageo's very first run said BUY at 2,475p; the fixes in rows 7, 9, 14 and 15 took it to HOLD at 1,627.3p before the final audit. The table shows each company before and after the evidence audit of 15 October.

Company	Before the audit	Final (15 Oct 2026) vs share price	What moved it
Diageo	HOLD 1,627.3p	HOLD 1,780.2p vs 1,673p	ERP 4.2%, diluted share count, FY30-31 growth 2.0%, exact beta (rows 27, 29, 33)
Coca-Cola	SELL USD 47.60	SELL USD 50.25 vs USD 90.12	ERP 4.2%, stakes not grown twice, bear margin (rows 27, 28)
Unilever	SELL 3,855.3p	HOLD 4,258.5p vs 4,710.5p (borderline)	ERP 4.2%, growth on the same businesses, pension deficit (rows 27, 30, 32)
RELX	SELL 1,621p	SELL 1,821p vs 2,771p	ERP 4.2% (row 27)

What is still not verified at source

Three inputs could only be seen second-hand, because the sites that publish them block automated access from the desk. They are disclosed in each memo: the UK gilt yield of 5.45% and the German Bund yield of 3.45% (aggregator websites), and EABL's sales (press reports). Everything else in the four memos traces to a filing, a data file or a database series. The 4.2% equity risk premium was also second-hand at first; on 16 Oct 2026, once the PM allowed the desk to reach Damodaran's site, it was confirmed at source (4.20% on 1 Oct 2026).

What I would tell another analyst using AI

Use the AI for reading, drafting and challenging, and keep the arithmetic in code you can test. Treat every number it proposes as a claim that needs a source, especially the ones that sound reasonable. And make the agents check each other: between them, the Bear case and Data agents caught four errors, against one found first by the independent auditor.